



Model Comparison Report

Overview

This report provides a comprehensive comparison of all trained deep learning models developed for skin disease classification using the HAM10000 dataset. It evaluates each model based on multiple performance metrics and identifies the best-performing architecture for accurate and reliable diagnosis.

Report Contents

1. Project Information

- Dataset: HAM10000 Skin Disease Dataset
- Total Classes: 7
- Image Size: 224 × 224
- Framework: TensorFlow / Keras
- Evaluation Date
- Total Models Compared

2. Models Evaluated

- Basic CNN
- Deep CNN
- CNN + Batch Normalization
- CNN + Dropout
- MobileNetV2
- EfficientNetB0
- ResNet50
- DenseNet121

3. Performance Comparison

Include a comparison table containing:

Model	Accuracy	Precision	Recall	F1 Score
Basic CNN	58.08%	74.50%	58.08%	62.89%
Deep CNN	53.42%	70.17%	53.42%	58.65%
CNN + BatchNorm	56.28%	67.68%	56.28%	60.12%
CNN + Dropout	45.01%	69.16%	45.10%	51.29%
MobileNetV2	56.75%	73.80%	56.75%	61.20%
ResNet50	66.93%	44.80%	66.93%	53.67%
EfficientNetB0	56.75%	68.72%	54.89%	56.67%
DenseNet121	57.35%	74.69%	57.35%	62.58%

4. Performance Visualizations

Include the following figures:

- Accuracy Comparison Chart
- Precision Comparison
- Recall Comparison
- F1 Score Comparison
- ROC Curves
- Precision–Recall Curves
- Training & Validation Accuracy
- Training & Validation Loss

5. Confusion Matrix

Display the confusion matrix for the selected or best-performing model to analyze class-wise prediction performance.

6. Hyperparameter Summary

Document the training configuration:

- Optimizer
- Learning Rate
- Batch Size
- Number of Epochs
- Data Augmentation
- Batch Normalization
- Dropout
- Early Stopping
- Learning Rate Scheduler

7. Best Performing Model

Summarize the top-performing model:

- Model Name
- Final Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Training Time
- Key Advantages

8. Observations

Highlight key findings, such as:

- Performance improvements achieved through optimization.
- Comparison between custom CNNs and transfer learning models.
- Impact of data augmentation and fine-tuning.
- Overall model stability and generalization.

9. Conclusion

Provide a concise conclusion, for example:

Among all evaluated architectures, the best-performing model achieved the highest classification accuracy while maintaining strong precision, recall, F1-score, and ROC-AUC. The combination of transfer learning, data augmentation, and model optimization significantly improved classification performance, making the selected model suitable for accurate skin disease diagnosis.

10. Report Metadata

Include:

- Report Generated On
- Project Name
- Framework Used
- Author
- Model Version